

The axis expansion of a one-variable amplitude

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Abstract

For an amplitude $\Psi(u, s)$ depending on the first coordinate only, Lipschitz in u at 0, the one-variable density of unit 86 expands as $\rho_\Psi(r, s) = r^{p-1} \log(1/r) \frac{\Psi(0, s)}{k_1 k_2} + r^{p-1} \left(\frac{\log R \Psi(0, s)}{k_1 k_2} + \frac{I_\Psi(s)}{k_2} \right) + O(r^{p-1+1/k_1} (1+s) e^{\beta L s})$ with the axis integral $I_\Psi(s) = \int_0^b (\Psi(u, s) - \Psi(0, s)) / u \, du$, and the density-transfer theorem gives $Z_\Psi(N) = N^{-p} (A_\Psi \log N + B_\Psi) + O(N^{-(p+1/k_1)} (1 + \log N))$ for $N \geq 1$. The constant B_Ψ sees the whole axis $\{v = 0\}$, not only the corner. Zero `sorry/axiom`.

1 The things formalised

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noncomputable def axisIntegral (b : ℝ) (Ψ : ℝ → ℝ) (s : ℝ) : ℝ :=
  u in Ioc (0 : ℝ) b, u ↦ Ψ u s - Ψ 0 s
theorem uDensity_integral_eq ... (m : ℝ) (hm : 0 < m) (hmb : m ≤ b) :
  ( u in Icc m b, u ↦ Ψ u s )
  = Ψ 0 s * Real.log (b / m) + axisIntegral b Ψ s
  - u in Ioc (0 : ℝ) m, u ↦ Ψ u s - Ψ 0 s
theorem uDensity_expansion ... :
  DensityExpansion (uDensity p b k k Ψ) ![p, p] ![1, 0]
  ![uCoeff k k Ψ, uCoeff b k k Ψ] (p + (k : ℝ) ^ 1) (b ^ (k + k)) 1
  (uEnvelope b L C k k)
noncomputable def axisA (p : ℝ) (k k : ℝ) (Ψ : ℝ → ℝ) : ℝ :=
  1 / ((k : ℝ) * k) * logMoment p 0 (Ψ 0)
noncomputable def axisB (b p : ℝ) (k k : ℝ) (Ψ : ℝ → ℝ) : ℝ :=
  1 / ((k : ℝ) * k) * (Real.log (b ^ (k + k)) * logMoment p 0 (Ψ 0)
  - logMoment p 1 (Ψ 0)) + 1 / (k : ℝ) * logMoment p 0 (axisIntegral b Ψ)
theorem twoDamp_uOnly_expansion ... :
  K' : ℝ, N : ℝ, 1 ≤ N →
  |twoDamp b N h h k k (fun u _ s => Ψ u s)
  - N ^ (-p) * (axisA p k k Ψ * Real.log N + axisB b p k k Ψ)|
  K' * N ^ (-(p + (k : ℝ) ^ 1)) * (1 + Real.log N)

```

2 Proof strategy

Split $u^{-1}\Psi(u, s) = u^{-1}\Psi(0, s) + u^{-1}(\Psi(u, s) - \Psi(0, s))$ on $[m, b]$, $m = (r/b^{k_2})^{1/k_1}$: the first part integrates to $\Psi(0, s) \log(b/m)$ with $\log(b/m) = (\log R + \log(1/r))/k_1$, the second is $I_\Psi(s)$ minus the tail over $(0, m]$, bounded by $m C_1(1+s)e^{\beta L s}$ since the integrand is bounded by $C_1(1+s)e^{\beta L s}$. Hence the density expansion with a degree-one log term and remainder r^{p-1+1/k_1} . The moments are finite by the log-power version of the polynomial-exponential bound lemma, and the transferred terms are $N^{-p}(\log N \cdot M_0 - M_1)$ (degree 1) and $N^{-p}M_0$ (degree 0), which rearrange to $A_\Psi \log N + B_\Psi$.

3 Roadblocks and resolutions

- Joint measurability of $(r, s) \mapsto \int_{m(r)}^b u^{-1} \Psi(u, s) du$: write the set integral as an integral of an indicator over $\mathbb{R}$ and use `StronglyMeasurable.integral_prod_right'` with the indicator function named explicitly (otherwise the typeclass search for the codomain is stuck).
- `Integrable.const_mul` of an `IntegrableOn` returns an `Integrable` on the restricted measure; ascribe the `IntegrableOn` type before `.congr_fun`.
- `logMoment.const_mul` rewrites also through eta-reduced $\Psi 0$; state the coefficients with $(\Psi 0)$ rather than `fun s =>  $\Psi 0 s$` .
- The envelope must be nonnegative for all s (structure field): use $(1 + |s|)$.

4 Outcome

The axis contribution of the $d = 2$ theorem is formalised. Remaining for the milestone: the anchored decomposition $F = F(u, 0) + F(0, v) - F(0, 0) + uvG$ with the mixed-term bound via a shifted constant-kernel block integral, and the assembly $Z(N) = N^{-p}(A \log N + B) + O(N^{-(p+\delta)}(1 + \log N))$.